



Genesis Biotech Inc.

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Hotstart KlenTaq DNA polymerase

Catalog No. PG-30006

Quantity: 250 units/ 50ul

Storage: -20°C

Description

Hotstart KlenTaq DNA polymerase is a modified version of KlenTaq DNA polymerase with hot start capability. Hotstart KlenTaq DNA polymerase is formulated with heat labile monoclonal antibodies that, at room temperature, effectively neutralize DNA polymerase activity. Full enzyme activity is regained upon denaturation of the antibody during the initial denaturation step. Using Hotstart KlenTaq DNA polymerase, hot start step is easily incorporated into PCR protocols already optimized with KlenTaq DNA polymerase, with little or no modification of cycling parameters or reaction conditions.

Storage buffer

This enzyme is supplied in 20 mM Tris-HCl, pH 8.0, 100 mM KCl, 0.1 mM EDTA, 0.5 mM DTT, 1% Triton X-100, 50% glycerol.

10X PCR buffer (1.0ml/vial)

The PCR buffer is supplied as a 10X concentrate and should be diluted for use.

Application

1. DNA amplification via PCR
2. Amplification of genomic DNA (complex templates)
3. Hot start PCR (polymerase is inhibited at room temperature during the reaction set up)
4. TA cloning
5. Realtime PCR application

Basic PCR Protocol

Add the following components to a sterile 0.5ml microcentrifuge tube sitting on ice:

Components	Volume	Final Concentration
10X PCR buffer (with MgCl ₂)	10 ul	1x
2.5mM dNTP	8 ul	200 uM
forward primer	1-5 ul	0.1-0.5 uM
reverse primer	1-5 ul	0.1-0.5 uM
KlenTaq DNA polymerase (5U/ul)	1 ul	5-6 unit
Template < 0.5 ug	0.05-0.5 ug for genomic DNA 0.5-50 ng for plasmid	

Sterile distilled water up to 100 ul.

Unit definition

One unit is the amount of enzyme required to catalyze the incorporation of 10 nmol of dNTP into acid-insoluble material in 30 minutes at 74°C, pH9.3, with activated salmon sperm DNA as the template-primer. The reaction conditions are: 25 mM TAPS (pH 9.3 at 25°C), 1 mM 2-mercaptoethanol, 2 mM MgCl₂, 50 mM KCl, 200 uM each of dATP, dTTP, dGTP, 100 uM [α -P³²]dCTP, 0.25 mg/ml activated salmon sperm DNA.

Purity

Nicking activity, endonuclease and exonuclease activity were not detected after the incubation of 0.6 ug of Lambda DNA, 0.6 ug of Lambda-HindIII digest or 0.6 ug of supercoiled pUC18 DNA, with 10 units of this enzyme for 1 hour at 74°C.

Related products

Taq DNA polymerase	PG-30001
Hotstart Taq DNA polymerase	PG-30002
Pr Taq DNA polymerase (Taq+Pfu)	PG-30003
Hotstart Pr Taq DNA polymerase	PG-30004
Klen Taq DNA polymerase	PG-30005
Real-time PCR kit	PG-30009
One Step RT-PCR kit	PG-30010
Genotyping kit	PG-30012
PCR kit	PG-30013